

CAMPUS WASTE SEGREGATION ASSISTANT

A Python Command-Line Project for Smarter Waste Disposal on Campus

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The Problem

Improper waste segregation on college campuses mixes recyclable, organic, electronic, hazardous, and general waste together. Students are often unsure how a specific item should be disposed of — and there's no quick, centralized way to check at the point of disposal.

Without a quick reference, campuses fall back on treating everything as general waste — losing most of its recycling and composting value.

1

Mixed Waste

Recyclables, organics, and hazardous items end up in the same bin.

2

Uncertainty

Students don't know which category uncommon items belong to.

3

No Visibility

No easy way to track what's being disposed of, or how often.

What It Does

Seven menu-driven features, all from a single terminal session.

1

Identify Waste

Get an item's category, bin, and disposal action.

2

View Categories

Browse all six categories with example items.

3

Search Waste Item

Flexible, case-insensitive keyword search.

4

Record Disposal

Save a dated entry to local JSON storage.

5

View Statistics

See totals and counts per category.

6

Environmental Tips

Practical waste-reduction guidance.

7

Disposal History

Review every past disposal record.

Six Waste Categories

Configurable to match any campus's actual bin colors and collection rules.

Organic

Food scraps, peels, leaves

GREEN / ORGANIC COLLECTION

Recyclable

Paper, cardboard, plastics

BLUE / RECYCLING COLLECTION

E-waste

Phones, chargers, keyboards

DESIGNATED E-WASTE POINT

Hazardous

Batteries, chemicals

HAZARDOUS-WASTE CHANNEL

Glass

Bottles and jars

GLASS COLLECTION POINT

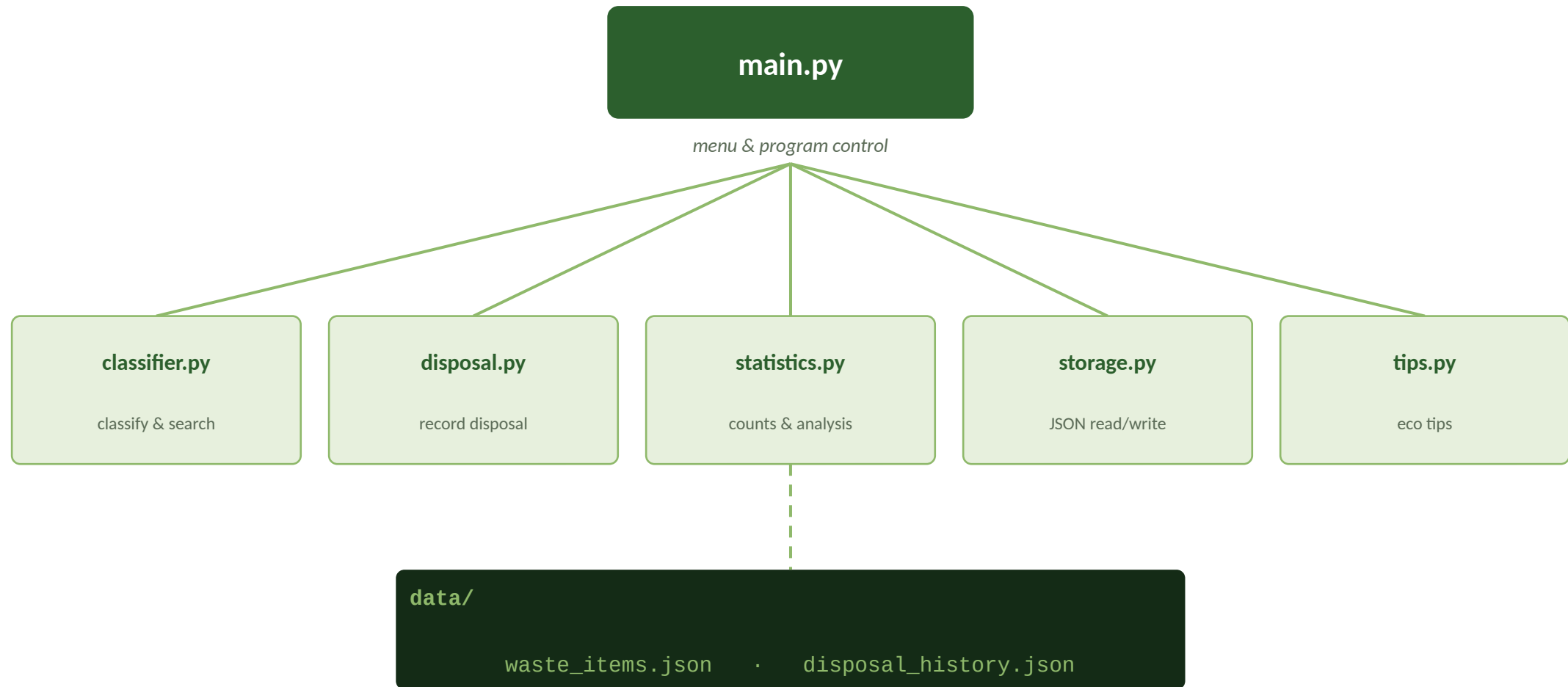
General

Everything else

GENERAL / REJECT CHANNEL

How It's Built

A small, modular codebase — one file per responsibility.



Under the Hood

Zero external dependencies. Runs anywhere Python 3 runs.

3.7+

Python version required

0

External dependencies

2

Local JSON data files

100%

Works fully offline

Python Concepts Demonstrated

Variables & data types

Functions & modules

Dictionaries & lists

File handling

Exception handling

JSON serialization

Input validation

String processing

Searching algorithms

Tested & Verified

Every case below was run manually in a live terminal session.



Valid item

“banana peel” → Organic



Keyword match

“old mobile phone” → E-waste



Empty input

Validation prompt, no crash



Disposal history

Records saved & displayed



Case variation

“PLASTIC BOTTLE” matches “plastic bottle”



Unknown item

Safe fallback message shown



Invalid menu choice

“99” → handled gracefully



Statistics

Category counts correct

8 / 8 test cases passed — including empty input, invalid menu choices, and unrecognized items.

Limitations & Future Scope

Current Limitations

- Bin names & rules must be configured per campus — nothing is universal.
- Only recognizes items in (or close to) its database.
- Data stored locally; not shared across devices or users.
- Command-line only — no graphical interface yet.

Future Scope

- Graphical or web-based interface.
- Multi-user accounts with cloud storage.
- Admin panel to edit the waste database.
- Barcode / image-based item recognition.

Get Started

Clone it, configure it to your campus, and run it from any terminal.

```
git clone https://github.com/YOUR-USERNAME/campus-waste-segregation-assistant.git
cd campus-waste-segregation-assistant
python main.py
```

Thank You

Built to make responsible waste segregation a little easier, one item at a time.